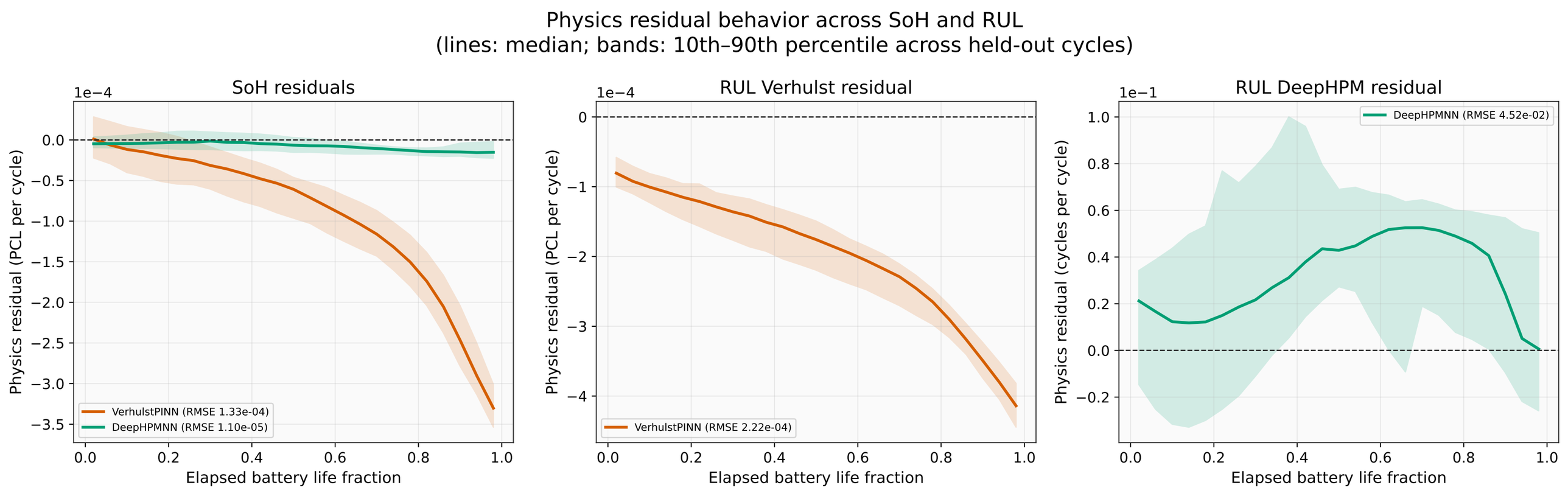
Physics residual behavior across SoH and RUL
(lines: median; bands: 10th–90th percentile across held-out cycles)
SoH residuals
1e−4
Physics residual (PCL per cycle)
Elapsed battery life fraction
VerhulstPINN (RMSE 1.33e-04)
DeepHPMNN (RMSE 1.10e-05)
RUL Verhulst residual
1e−4
Physics residual (PCL per cycle)
Elapsed battery life fraction
VerhulstPINN (RMSE 2.22e-04)
RUL DeepHPM residual
1e−1
Physics residual (cycles per cycle)
Elapsed battery life fraction
DeepHPMNN (RMSE 4.52e-02)